

BUILD SPEC — SamInnova

Portfolio

This file explains HOW the site should be built: stack, design system, pages, and the admin panel. Pair it with `01 - PROJECT-CONTEXT.md` (the facts/content rules).

Tech stack (locked)

Layer	Technology
Frontend structure	HTML5
Styling	CSS3 (custom — no heavy framework)
Interactivity/animation	Vanilla JavaScript
Projects section (dynamic)	PHP
Database	MySQL
Admin panel	PHP + MySQL, session-based auth
Hosting	AlwaysData

Layer	Technology
Deployment	FileZilla / WinSCP
Version control	Git / GitHub

Static pages (plain HTML/CSS/JS, no database): Home, About, Skills, Experience, Contact

Dynamic (PHP + MySQL): Projects section + everything the admin panel edits

Why this split: keeps most of the site fast/simple, while Projects demonstrates real full-stack + database ability — relevant for a MIAAGE grad's recruiter audience.

Language

Bilingual FR/EN. Toggle in the nav, persists via `localStorage`. All static content needs both languages written out. Database-driven project fields also need `_fr` / `_en` columns (see schema below).

Design system (already established — reuse exactly, don't reinvent per page)

Aesthetic: Dark tech/developer theme, restrained — not neon-green cliché, not generic template.

Colors

- `--bg : #0A0F1C` (deep ink navy, not pure black)
- `--surface : #121A2C`
- `--surface-2 : #182238`
- `--line : #24304A`
- `--text : #E9EDF6`
- `--text-dim : #8D97AE`
- `--accent : #E8A33D` (warm amber)
- `--accent-dim : #B9822E`

Typography

- Display/headings: **Space Grotesk**
- Body: **Inter**
- Labels/eyebrows/mono accents: **JetBrains Mono**

Signature visual motif: an animated SVG "node/schema" graphic (dots connected by lines, representing a systems/tech diagram) — ties to MIAGE/information-systems identity. Used in the hero; draws in on load, nodes gently float.

Motion requirements (site must feel "alive")

- Scroll-triggered reveals via `IntersectionObserver` (fade + slide left/right/up depending on section, staggered)
- Infinite auto-scrolling tech logo marquee, pauses on hover (uses devicon CDN icons)
- Subtle hover animations on cards/buttons (lift + border/glow)
- Everything must respect `prefers-reduced-motion` (disable animations)

Page-by-page structure

Home — hero (name, positioning line, value prop, CTA buttons), animated schema graphic, tech marquee, 2–3 featured project cards, CTA banner to About/Contact.

About — MIAGE journey (2023–2026), transition to SamInnova, personality (practical/self-directed/entrepreneurial), scroll-reveal per block, optional photo.

Projects (dynamic) — grid pulled from MySQL, filter tabs by category, each project links to a case-study view (problem/solution/tech/role/screenshots/live link/GitHub if they exist). Placeholder content until real project data is added (see content rules in file 01).

Skills — grouped: Frontend / Backend / Databases / Frameworks / Tools & Deployment. Icons/logos, not percentage bars (avoids overstating proficiency).

Experience/Journey — simple animated timeline: MIAGE start → key projects → SamInnova → present.

Contact — email, phone, LinkedIn, GitHub, CV download, contact form or mailto (decide backend approach at build time). Real verified info only (see file 01).

Admin panel spec

Scope: Projects only. Nothing else on the site is admin-editable.

projects table

```
id (PK)
title_fr, title_en
category (business site / booking system /
digital menu / django app / other)
description_fr, description_en
problem_fr, problem_en
solution_fr, solution_en
tech_stack (comma-separated or JSON)
role
screenshot_url
live_url (nullable)
github_url (nullable)
status (completed / near-completion / in-
progress) -- must reflect true status
display_order
created_at, updated_at
```

admin_users table

```
id (PK)
username
password_hash
last_login
```

Features: secure login

(`password_hash()` / `password_verify()` , PHP sessions, CSRF tokens, rate-limiting/lockout on failed logins), dashboard listing all projects with edit/delete, add/edit form (bilingual fields, image upload with validation, status dropdown), delete with confirmation, logout. All `/admin`

routes require valid session. All DB queries via prepared statements — no raw SQL concatenation, ever.

Accessibility requirements

Semantic HTML, proper heading hierarchy, alt text on all images, WCAG AA contrast, visible keyboard focus states, logical tab order, `prefers-reduced-motion` fallback, associated form labels/errors, responsive mobile/tablet/desktop.

Build sequence

1. Static HTML/CSS structure, all pages, mobile-first
2. Visual identity/base styles (done — see design system above)
3. Animation system (scroll-reveal, marquee, transitions)
4. Bilingual FR/EN toggle system
5. MySQL schema + PHP backend for Projects
6. Public Projects page pulling from DB
7. Admin panel: auth, dashboard, CRUD
8. Contact form wiring
9. Accessibility pass
10. Performance pass (image optimization, lazy loading, minification)

11. Deploy to AlwaysData